Data Drift in Machine Learning Systems

1. Definition

Data Drift refers to a statistically significant change in the distribution of input data (and sometimes target data) between model training time and production time. When drift occurs, model performance degrades because the model's learned patterns no longer align with real-world data.

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2. Types of Data Drift

Covariate Drift: Change in input feature distribution P(X). Prior Probability Drift: Change in target distribution P(Y). Concept Drift: Change in relationship between input and target P(Y|X).

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3. Why Data Drift Matters

- Silent degradation of model accuracy - Business KPI impact (revenue loss, false positives) - Regulatory and compliance risks - Loss of stakeholder trust

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4. Drift Detection Techniques

- Kolmogorov–Smirnov (KS) Test - Chi-Square Test (for categorical features) - Jensen-Shannon Divergence - Population Stability Index (PSI) PSI Interpretation: < 0.1 : No drift 0.1-0.25: Moderate drift > 0.25 : Significant drift

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5. Monitoring Architecture Pattern

Training -> Deploy -> Monitor -> Detect Drift -> Retrain -> Redeploy Typical components: - Data logging layer - Drift detection service - Alerting system - Automated retraining pipeline

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6. Mitigation Strategies

Short-Term: - Threshold-based alerts - Manual review - Rule-based overrides Medium-Term: - Scheduled retraining - Online learning Advanced: - Continuous training (CI/CD for ML) - Champion-Challenger models - Shadow deployments

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7. Executive-Level Summary

Data drift is inevitable in production ML systems. Organizations must implement continuous monitoring, statistical detection methods, automated retraining pipelines, and governance policies to maintain model reliability and business alignment.
